

Topics on sufficiency

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Theorem 1 (Factorization theorem: sufficiency direction). *Let X have density or probability mass function $f_\theta(x)$ with respect to a common dominating measure. Suppose there exist nonnegative functions g_θ and h such that*

$$f_\theta(x) = g_\theta(T(x)) h(x),$$

where h does not depend on θ . Then $T = T(X)$ is sufficient for θ .

Proof. We show that the conditional distribution of X given T does not depend on θ .

For intuition, suppose that after a suitable reparameterization we may write

$$X = (T, U),$$

where U denotes the remaining information in the sample after T is fixed. The joint density of (T, U) then has the form

$$f_\theta(t, u) = g_\theta(t) h(t, u).$$

The marginal density of T is

$$\begin{aligned} f_{T,\theta}(t) &= \int f_\theta(t, u) du \\ &= \int g_\theta(t) h(t, u) du \\ &= g_\theta(t) \int h(t, u) du. \end{aligned}$$

Therefore, whenever $f_{T,\theta}(t) > 0$, the conditional density of U given $T = t$ is

$$\begin{aligned} f_\theta(u \mid T = t) &= \frac{f_\theta(t, u)}{f_{T,\theta}(t)} \\ &= \frac{g_\theta(t) h(t, u)}{g_\theta(t) \int h(t, v) dv} \\ &= \frac{h(t, u)}{\int h(t, v) dv}. \end{aligned}$$

The final expression does not depend on θ . Hence the conditional distribution of the sample given T is independent of θ , which ensures sufficiency of T .

□

Theorem 2 (Lehmann-Scheffe theorem). *Let T be a complete and sufficient statistics for θ . If $\delta(T)$ is an unbiased statistics for $\tau(\theta)$, i.e.,*

$$\mathbb{E}[\delta(T)] = \tau(\theta) \quad \text{for all } \theta,$$

then $\delta(T)$ is the unique UMVUE for $\tau(\theta)$.

Proof. UMVUE property. We will first show that $\delta(T)$ is UMVUE. Let V be an unbiased estimator of $\tau(\theta)$, and define $\delta_V(T) = \mathbb{E}[V | T]$. By Rao-Blackwell, we already know that

$$\text{Var}(\delta_V(T)) \leq \text{Var}(V). \quad (1)$$

In addition, we have $\mathbb{E}[\delta_V(T)] = \tau(\theta) = \mathbb{E}[\delta(T)]$. Therefore, by completeness, we can conclude that

$$\mathbb{E}[\delta_V(T) - \delta(T)] = 0 \Rightarrow \delta_V(T) \stackrel{a.s.}{=} \delta(T).$$

.

Plugging the above in (1), we get

$$\text{Var}(\delta(T)) \leq \text{Var}(V) \quad \text{for any unbiased } V. \quad (2)$$

This shows that $\delta(T)$ is UMVUE.

Uniqueness. Let us assume that U is another UMVUE. Then $\delta_U(T) := \mathbb{E}[U | T]$ enjoys $\text{Var}(\delta_U(T)) \leq \text{Var}(U)$. However, U is UMVUE, which yields that $\text{Var}(U) \geq \text{Var}(\delta_U(T))$. Combining these facts we have $\text{Var}(\delta_U(T)) = \text{Var}(U)$. Due to law of total variation, we have

$$\text{Var}(U) = \text{Var}(\delta_U(T)) + \mathbb{E}[\text{Var}(U | T)] \implies \mathbb{E}[\text{Var}(U | T)] = 0.$$

This shows that $\text{Var}(U | T) = 0 \implies U = \mathbb{E}[U | T] = \delta_U(T)$. Therefore, again using completeness, we can conclude

$$\delta(T) = \delta_U(T) = U.$$

This finishes the proof. □

Theorem 3 (Minimal sufficiency in a full-rank exponential family). *Suppose that X has density or probability mass function*

$$f_\theta(x) = h(x) \exp \{ \eta(\theta)^\top T(x) - A(\theta) \}, \quad \theta \in \Theta,$$

with respect to a common dominating measure, where

$$T(x) = (T_1(x), \dots, T_k(x))^\top.$$

Assume that the natural parameter space

$$\mathcal{H} = \{ \eta(\theta) : \theta \in \Theta \}$$

contains a nonempty open subset of $\mathbb{R}^k$. Then $T(X)$ is a minimal sufficient statistic for θ .

Proof. By the factorization theorem,

$$f_{\theta}(x) = \underbrace{\exp \{ \eta(\theta)^{\top} T(x) - A(\theta) \}}_{g_{\theta}(T(x))} h(x),$$

so T is sufficient.

To establish minimal sufficiency, we use the likelihood-ratio criterion. For two sample points x and y with $h(x), h(y) > 0$,

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} = \frac{h(x)}{h(y)} \exp \{ \eta(\theta)^{\top} (T(x) - T(y)) \}.$$

If $T(x) = T(y)$, then

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} = \frac{h(x)}{h(y)},$$

which is independent of θ .

Conversely, suppose that

$$\frac{f_{\theta}(x)}{f_{\theta}(y)}$$

is independent of θ . Then

$$\eta^{\top} (T(x) - T(y))$$

must be constant as a function of η for all $\eta \in \mathcal{H}$.

Let

$$d = T(x) - T(y).$$

Thus $\eta^{\top} d$ is constant on an open subset of $\mathbb{R}^k$. Choose an interior point η_0 of this subset. Since the set is open, for every sufficiently small $v \in \mathbb{R}^k$,

$$\eta_0 + v \in \mathcal{H}.$$

Hence

$$(\eta_0 + v)^{\top} d = \eta_0^{\top} d.$$

Therefore

$$v^{\top} d = 0$$

for every sufficiently small $v \in \mathbb{R}^k$.

Taking $v = \varepsilon e_j$, where e_j is the j th standard basis vector and ε is sufficiently small, gives

$$\varepsilon d_j = 0,$$

and hence

$$d_j = 0, \quad j = 1, \dots, k.$$

Thus

$$d = 0,$$

so

$$T(x) = T(y).$$

We have therefore shown that

$$T(x) = T(y) \iff \frac{f_{\theta}(x)}{f_{\theta}(y)} \text{ is independent of } \theta.$$

By the likelihood-ratio criterion for minimal sufficiency, $T(X)$ is minimal sufficient for θ . $\square$